LECTURE 17

SUMMARY

MODULE 3 –LECTURE 3

How does muscle contract

TYPES OF HYPERTROPHY

MUSCLE HYPERPLASIA

Increase in the quantity of muscle cells, very limited evidence of it being possible among humans, not a primary contributor to growth.

MYOFIBRILLAR HYPERTROPHY

Fibre of the muscle is called myofibrillar increase in the size of the fluid that surrounds your contractile elements of the skeletal muscle; inflation in the muscle cell
It includes those workouts which is tension based with lower rep range, overload of the muscles.

Increases the density of the fibres of the muscles

SARCOPLASMUC HYPERTROPHY

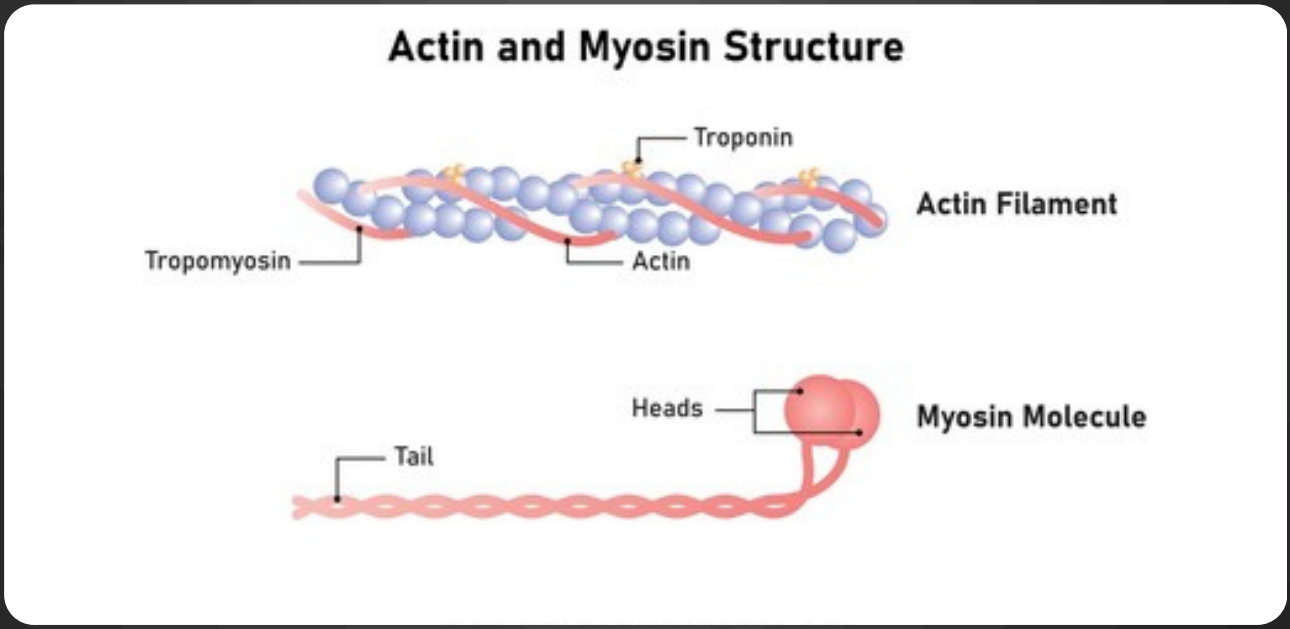
changes in the sarcoplasm, presence of all of the metabolic enzymes like ribosomes, t- tubules etc which help the contractile elements to
actually move, improved metabolic function
It includes metabolic stress based workouts like HIIT, functional training etc.

Increase in the fluid to improve cell metabolic health

However, it is possible that a sarcoplasmic workout is also providing the myofibrillar hypertrophy.

HOW DOES A MUSCLE CONTRACT/ PRODUCE TENSION?

- Tension is the most important thing that a muscle needs.
- Actin & Myosin: responsible for muscle shortening or lengthening when the muscle shortens, actin & myosin bind together.



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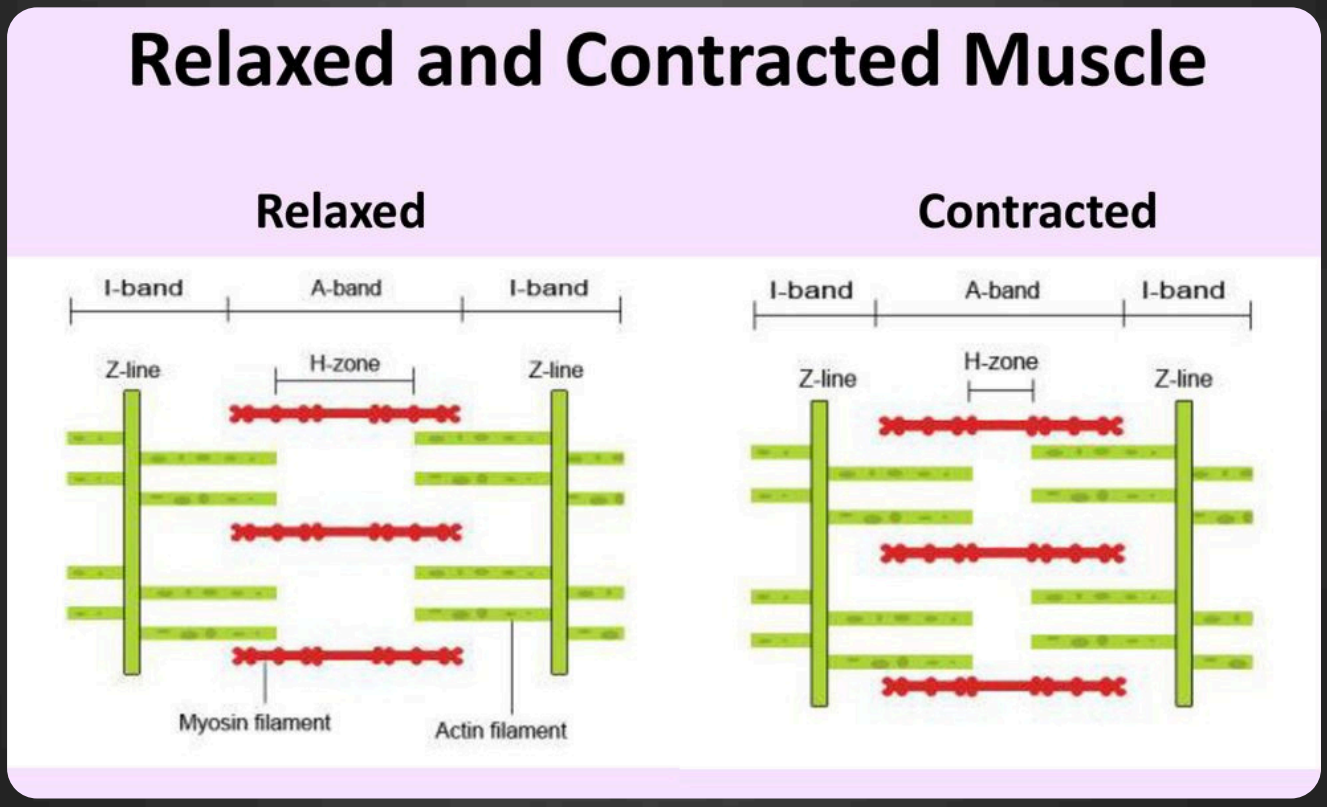
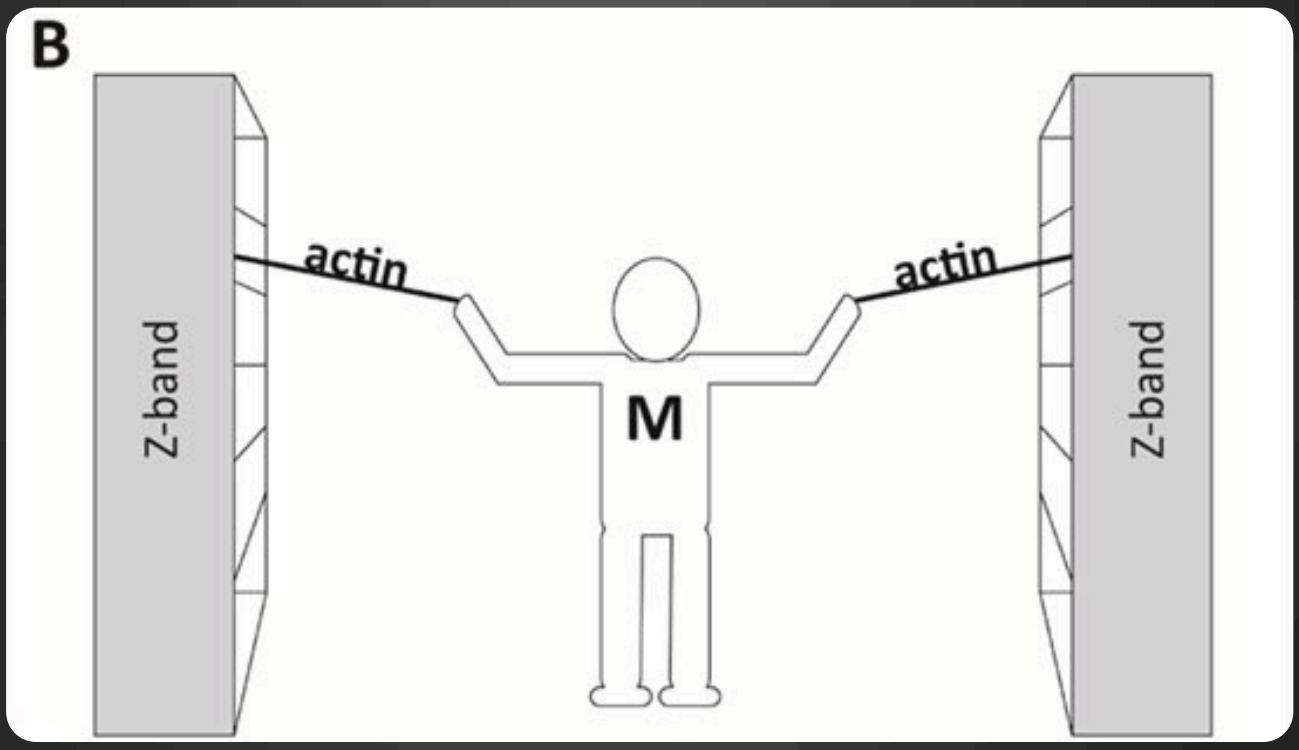
SLIDING FILAMENT THEORY

The sliding filament theory explains the mechanism of muscle contraction based on muscle proteins that slide past each other to generate movement.
According to the sliding filament theory, the myosin (thick filaments) of muscle fibers slide past the actin (thin filaments) during muscle contraction, while the two groups of filaments remain at relatively constant length.



SLIDING FILAMENT THEORY

1. Signal: Your brain tells the muscle to move.
2. Calcium: This signal releases calcium inside the muscle.
3. Attachment: Calcium helps myosin attach to actin.
4. Pull: Myosin pulls actin, making the muscle shorter and contract.
5. Release: Myosin lets go of actin when it gets new energy (ATP).
6. Reset: Myosin gets ready to pull again by using energy.
7. Relax: When the signal stops, calcium goes away, and the muscle relaxes.



The neural signal, to carry forward this process, is sent through a nerve which will innovate a certain amount of muscle cells. The number of cells involved depends on the threshold of the motor unit.

Motorunits defined as a motoneuron and all of its associated muscle fibres, are the basic functional units of skeletal muscle. Their activity represents the final output of the central nervous system.

High Threshold Motor Unit: Challenging Movements, failure based movements, effort demanding, more growth Low Threshold Movements: Easy Movements, tension based movements, fatigue resistance.

While training, high threshold motor unit is more optimal to recruit more muscle fibres at the same time.